Angled and negative work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use force direction to determine work's sign and size.

SUCCESS CRITERIA - I can...

1. Compare force direction with displacement.
2. Use the parallel force component.
3. Explain positive, negative and zero work.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A fixed impulse changes a target's sideways velocity by **0.03 m/s**. **Find the predicted shift after 500 s** and **state a limitation**.

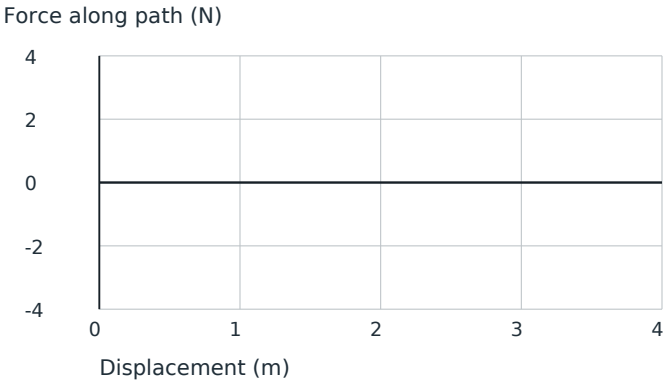
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $W = Fd\cos\theta = F_{parallel}\,d$.
θ is between force and displacement. CPA uses the supplied parallel component. Opposite force gives negative work.

Worked reasoning: A box moves **4 m** right while friction acts **3 N** left. Friction's work is **-3×4=-12 J** because force opposes displacement.

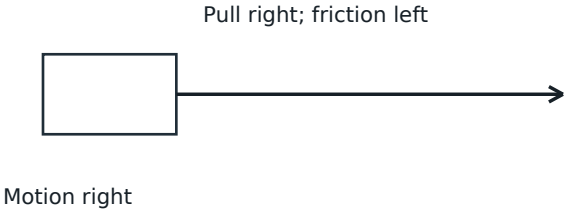
Your turn: 1. **Draw force** and displacement arrows.
2. **Add a perpendicular upward force** and **state its work** for this horizontal motion.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The box moves right. Classify work by a rightward pull, leftward friction and upward normal force.



The rectangle is the box. Add the requested force and displacement labels.

Marzano check - circle: ① ② ③

04 Apply the physics

A rope pulls a crate **3 m** right. Its forward force component is **8 N**; friction is **2 N** left.
1. **Draw the force components** and displacement.
2. **Calculate rope work**, friction work and net work.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A force perpendicular to velocity can bend a path without changing speed. Explain using work.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two carts separate from rest with no external horizontal impulse. One has momentum **24 kg·m/s** left. **Find the other's momentum**.
Use before and after snapshots to organize the momentum account.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A **5 N** friction force acts opposite a **4 m** displacement. **State work** with sign and units.

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A pull does **+20 J** and friction does **-8 J**.
1. **Draw work-transfer arrows** and **find net work**.
2. **What changes** if friction is twice as large over the same path?

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3